Main-line synthesis: the Reading-H vacuum-selection theorem assembled from the five published referee packages

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

This is the claim-level synthesis (task T-013): one self-contained map of how the published main-line theorem rests on its five PUBLISHED referee packages. It adds no new mathematics and enacts no tier change; every load-bearing statement below carries the bundle ID, tier, and scope under which it was operator-confirmed. **Scope:** the main proof line of the Reading-H full-class vacuum selection (B1/B2), with B5 supplying the off-diagonal machinery. **Not claimed:** anything beyond the composite statement of Sec. 2 – in particular no global / all-intensity / A1-free claim, no arbitrary non-lattice DR-2, and no TECT TOE completion.

2. The theorem being synthesised

Theorem (Reading-H selection over $\mathcal{C}_{\text{full}}$; enacted T7-SCOPE). *Within the certified window $\mathcal{R} = \{\mu^2 \in (0, 0.0342),\ I \in (0, I_c^{\text{sel}}(\mu^2))\}$, given the A1 kernel convention, every admissible competitor $Q \in \mathcal{C}_{\text{full}}$ (real antipodal pattern, $|Q| \le n_{\text{pack}} = 16/\theta_{\min}^2$, pairwise geodesic separation $\ge \theta_{\min}$) satisfies*

$$F[Q] - F[G_*] \ > \ 0.$$

The proof has the three-sector structure (D)(O)(S) of the Reading-H package: (D) the block-diagonal isotropy Hessian (entropy floor $\frac{1}{2}r_R^2 > 0$, breathing $\frac{3}{2}u_{\text{eff}} > 0$), competitor-agnostic; (O) the off-diagonal bound $R_{\text{lead}} \le 23.2 \cdot 11 \cdot I \le 0.510 < 1$ ($\times 1.96$ headroom) from Lemma 1 ($K_{\text{floor}} \le T'$) and Lemma 2 (coherence circle-packing, $T' \le \lfloor 2\pi/\theta_{\min} \rfloor = 10$); (S) the third-cumulant selection floor, joint $= \times 1.04 > 1$ at the endpoint.

3. The dependency DAG: five published packages, per-edge tier and scope

Package	Bundle	Tier	Scope	Role (edge into the theorem)
Reading-H C_{full}	`Reading-H-cFull-T7-260611`	T7-SCOPE	$\mathcal{C}_{\text{full}}$, window $\mathcal{R}$, given A1	the head: (D)(O)(S) assembly + window certification (res5_032–036)
Prop-A	`Prop-A-T6-260611`	T6	$\mathcal{A}_{\text{adm}}$ (registered shells, $T' \leq 13$), operating point	(D) strict comparison infimum + (O) class-wide closure on the primary class; sub-structure of the head ($\mathcal{A}_{\text{adm}} \subset \mathcal{C}_{\text{full}}$)
DR-2 lattice	`DR2-Lattice-T7-NTstandard-260612`	T7 (lattice; std. NT import)	$Q = \Lambda \cap \{\lvert x\rvert^2 = R\}$	Lemma 1 backing: $E_+ \leq_\varepsilon R^\varepsilon N^2$ decoupling-free; the lattice-class confirmation of the sum-circle mechanism
STEP-5B rectangle	`STEP-5B-Rectangle-T6-260612`	T6	rectangle prefactor + official threshold	the K-budget: $K(n) \leq 8 + 4\sqrt{14}\,\kappa^4\sqrt{n}$, $n_{\text{adm}} \lesssim 1.59 \times 10^5$ at the anchor
SC-SCOPE	`SC-SCOPE-T5-260612`	T5 thin	endpoint $I = 2 \times 10^{-3}$ + window $\mathcal{W}_{\text{SC}}$	(S): joint $= \times 1.040$–$\times 1.082 > 1$, sunset-limited

Named definitional input: A1-KERNEL-CONV ($r_{\text{braz}} = K(q_0) = \mu^2$), the microscopic-theory convention every package assumes. Numerical base layer: B1-enumerated (T5 CLOSED@ESTIMATOR-GRADE, 167/167 asserts) – corroborating, not load-bearing for the analytic chain. The DAG is acyclic; per policy sec.15 the supports were published before this capstone.

4. Composite-grade accounting (honest tier algebra)

The head label T7-SCOPE$_{C_{\text{full}}}$ was enacted by the Reading-H package itself (operator-confirmed); this note verifies that the composition is sound:

(a) Where the T5/T6 supports enter. (D) and (O) are proved competitor-agnostically IN the head package; Prop-A (T6) strengthens (D)(O) on $\mathcal{A}_{\text{adm}}$ and DR-2 (T7) confirms Lemma 1's mechanism on the lattice class – neither weakens the head. The two genuinely load-bearing imports below T7 are: (i) **(S) at T5 thin**: the selection floor closes at $\times 1.04$–$\times 1.08$, sunset-limited, certified on $\mathcal{W}_{\text{SC}}$ only – the head inherits this thinness at the endpoint and says so; (ii) **the K-budget at T6**: the rectangle threshold 1.59×10^5 vastly clears the class size ($n_{\text{pack}} \approx 44$–$49$): $K(n_{\text{pack}}) \leq 8 + 4\sqrt{14}\,\kappa^4\sqrt{49} \approx 113$, against $K_{\text{budget}} = 5972$ at the anchor ($\times 53$; the measured value $K = 107$, margin $\times 55.6$, cross-checks the bound at $\times 1.06$ accuracy).

(b) Why arbitrary-Q DR-2 is NOT load-bearing. Within $\mathcal{C}_{\text{full}}$, Lemma 2 (coherence circle-packing) caps the sum-circle occupancy ELEMENTARILY: $T' \leq \lfloor 2\pi/\theta_{\min} \rfloor = 10$. The admissibility constraints ($\theta_{\min}$ separation) do the work that unrestricted additive-energy bounds would otherwise have to do; the Bourgain–Demeter-conditional arbitrary-Q route is a FRONTIER STRENGTHENING (removing the admissibility cap), not a gap in the present theorem.

(c) Weakest-link list (explicit). A1 (definitional input, not proved here); (S) thinness $\times 1.04$ at the endpoint; the window $\mathcal{R}$ / $\mathcal{W}_{\text{SC}}$ boundaries; Prop-A's full-$\mathcal{C}_{\text{full}}$ extension running through the head package rather than independently. Any falsifier listed in the five footers falsifies the composite.

5. Track separation: closed vs genuinely remaining

CLOSED by the five packages: the main-line theorem of Sec. 2 at its enacted scope; the K-budget threshold; the lattice-class DR-2; the endpoint selection floor; reproduction: 27 entry scripts / 330 asserts across the five bundles, all operator-clean-run-confirmed, derived from the bundle MANIFESTs as

$$\underbrace{5}_{\text{RH}} + \underbrace{9}_{\text{PA}} + \underbrace{8}_{\text{DR2}} + \underbrace{1}_{\text{5B}} + \underbrace{4}_{\text{SC}} = 27 \text{ scripts}, \qquad \underbrace{27}_{\text{RH}} + \underbrace{44}_{\text{PA}} + \underbrace{38}_{\text{DR2}} + \underbrace{192}_{\text{5B}} + \underbrace{29}_{\text{SC}} = 330 \text{ asserts.}$$

REMAINING (separate tracks, registered in the task ledger; none blocks Sec. 2):

- **T-014 arbitrary-Q DR-2** (frontier; T6-conditional on Bourgain–Demeter): would remove the admissibility cap from Lemma 1's backing. Research-grade open problem; not load-bearing (Sec. 4b).
- **T-015 full STEP-5B closure**: the budget COMPARISON is machine-closed ($\times 55.6/\times 8.8/\times 2.1$–$2.6$ across the band); what remains is the admissible-class EXHAUSTIVENESS decision layer (H-ADM-COH adoption record; operator-decision items per the B5 ledger), plus the named backlog lemma R-U6-1 (T-004, tadpole alignment).
- **Beyond this synthesis**: per-claim SYNTHESIS layers for B1/B2/B5 individually (T-013 continuation, one note per turn); the claim-card tier reviews that may follow from the published packages (operator-decision).

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"A synthesis can over-claim by composition: T5 inputs under a T7-SCOPE head."* ADDRESSED: Sec. 4a names the only two load-bearing sub-T7 imports and shows where their weakness lands (endpoint thinness; T6 threshold with $\times 53$ clearance). The head label is the operator-enacted one; this note's accounting narrows, never widens, the claim.

(β) *"The DAG could hide a cycle (head cited by a support)."* DISMISSED: the five bundles were published in dependency order (sec.15); each support's footer cites only sub-results or definitional inputs, never the head; verified by reading the five footers (Sec. 3 table).

(γ) *"Lemma 2's $T' \leq 10$ and Prop-A's $T' \leq 13$ scopes differ – is the (O) chain consistent?"* VALID with clarification: Lemma 2's cap is the $\mathcal{C}_{\text{full}}$ admissibility cap used by the head's (O); Prop-A's $T' \leq 13$ is the registered-shell class $\mathcal{A}_{\text{adm}}$ bound used for its own (O) closure. Both chains close independently with their own constants; no constant is shared across the two scopes.

Quantitative sanity check (mandatory). Cross-package consistency: $K(n_{\text{pack}})$ predicted by the STEP-5B rectangle bound (≈ 113 at $n = 49, \kappa \simeq 1$) vs the measured $K = 107$ – agreement to 6%, bound on the safe side; $R_{\text{lead}} = 0.510 < 1$ with $\times 1.96$; joint $\geq 1.040 > 1$; $n_{\text{pack}} \approx 49 \ll 1.59 \times 10^5$ ($\times 3200$ clearance); all five bundle digests verified at publication (sha256, per-file).

7. Result footer

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Result ID:                Main-Line-Synthesis-T013-260612
                          (PUBLISHED SYNTHESIS NOTE) -- B1-RH-ENUM /
                          B2-PROPA-HLAYER / B5-BEYOND-LAYER-BOUND
Precise statement:        The Reading-H selection theorem (Sec. 2;
                          T7-SCOPE_{C_full} given A1, window R) is
                          assembled from five PUBLISHED packages:
                          Reading-H-cFull-T7-260611 (head),
                          Prop-A-T6-260611 (D/O on A_adm),
                          DR2-Lattice-T7-NTstandard-260612 (Lemma 1,
                          lattice), STEP-5B-Rectangle-T6-260612
                          (K-budget, 1.59e5), SC-SCOPE-T5-260612
                          ((S) endpoint floor). DAG acyclic;
                          composite grade = the head's enacted
                          T7-SCOPE with (S) thinness and endpoint
                          window inherited and flagged.
Scope:                    synthesis layer only -- no new mathematics,
                          no tier change, no gate flip. Tracks T-014
                          (arbitrary-Q DR-2) and T-015 (STEP-5B
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	exhaustiveness decision layer) are registered as NOT load-bearing for Sec. 2.
Evidence grade:	SYNTHESIS over five operator-confirmed PUBLISHED bundles (27 entry scripts / 330 asserts = 5+9+8+1+4 / 27+44+38+192+29 from the bundle MANIFESTs, all clean-run confirmed; v1.2 corrected count).
Reproduction command:	run the five bundles' entry manifests (see each bundle's README); this note adds no new scripts.
Falsifier:	any falsifier of the five cited footers; or a demonstrated cycle / scope mismatch in the Sec. 3 DAG.
Tier before / after:	no change BY THIS NOTE (synthesis layer). Canonical states: B1 and B2 T7-SCOPE_{C_full} (joint head enactment); B5 T6 PROVED-CONDITIONAL on H_B5^T6 (re-tiered by T-031 D3-A, 2026-06-12; v1.2's 'B5 T5' was the pre-re-tier state and 'B2 T6' was a claim-vs-unit conflation, both fixed v1.3).
No-overclaim statement:	this note narrows the published claims by making every scope explicit; it claims nothing the five footers do not.
Next required action:	none for this note (PUBLISHED). Per-claim SYNTHESIS layers (B1/B2/B5) issued alongside this version; tracks T-030/T-031 per ledger.